

$U(1)$ Kac–Moody Level and Rényi Entanglement in Monitored Gaussian States

Class-A Fermionic Adaptive-Circuit Project
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We derive the relation between interval entropies, charge fluctuations, and the infrared data of a fermionic boundary theory. We first obtain exact Gaussian-state formulas from the restricted correlation matrix, keeping conditional observables distinct from those of a measurement-record mixture. We specify chiral and nonchiral fermion Hamiltonians and actions, explain their bosonization, and compare ordinary and chiral Luttinger liquids. We then explain the replica construction of the interval entropy and derive the cylinder charge variance from a normalized current algebra. We introduce the full affine Kac–Moody algebra, its representations, and its relation to the Virasoro algebra before specializing to physical charge. These calculations identify different quantities: entropy measures the sum of left- and right-moving central charges, while charge fluctuations measure the current levels. Their relation for a charge-one complex fermion is explained using the Sugawara construction and the common susceptibility spectrum. An interface-defect example explains how logarithmic entanglement can survive with a continuously varying coefficient and a gapped modular spectrum, without changing the bulk central charge or current algebra. Finally, we discuss one-wall versus two-wall normalization and the interpretation of spatial entanglement contours. The Gaussian identities are exact; their conformal interpretation requires additional assumptions about the long-distance state.

CONTENTS

I. Gaussian reduced states and interval entropies	2
A. Correlation convention and natural orbitals	2
B. Taking the trace of a power	3
II. Charge statistics and measurement-record averages	4
A. Generating function and cumulants	4
B. Which average is being taken?	4
III. Continuum fermions, bosonization, and Luttinger liquids	4
A. A Dirac fermion and a single chiral fermion	5
B. Bosonization conventions and the fermion dictionary	5
C. The ordinary Luttinger liquid	6
D. The chiral Luttinger liquid and its charge normalization	7
IV. Replica derivation of the interval entropy	7
A. The conformal ingredients	8
B. From gluing replicas to twist fields	8
C. Plane-to-cylinder map and the Rényi coefficient	9
V. The physical current and its Kac–Moody level	10
A. From a symmetry to the full affine algebra	10
B. The Abelian current, its states, and physical normalization	10
C. Density correlator on a circle	11
D. Integrating the correlator	11
E. A regulated mode sum: where the contact term went	12
VI. Why central charge and current level are distinct	12
A. Sugawara construction and charged descendants	12
B. Neutral sectors and the limits of coefficient matching	13
VII. A common susceptibility spectrum	13
A. Exact spectral relations	13
B. Modular-energy kernels and their integrals	14

C. Why a homogeneous critical chain can have a flat coefficient	15
VIII. A solvable interface defect: logarithms without homogeneous coefficients	15
A. The spectral map and the logarithmic mode count	15
B. The entropy integral and effective central charge	16
C. Susceptibility, charge fluctuations, and the nonflat density	17
IX. What can be inferred about individual walls?	17
A. One-wall and two-wall normalization	17
B. Where chirality appears, and why the two-wall total cancels	18
C. Entanglement contours and wall windows	19
X. Scope of the derivations	19
References	20

I. GAUSSIAN REDUCED STATES AND INTERVAL ENTROPIES

The purpose of this note is to connect two descriptions of the same subsystem. Microscopically, a number-conserving Gaussian state is specified by its two-point function. At long distances, a critical interface may instead be described by a conformal field theory (CFT). We will derive the microscopic entropy and charge observables first, and only then introduce the assumptions under which their logarithmic coefficients become CFT data. We use the correlation-matrix convention of Bhuiyan, Pan, and Jian (BPJ) [1].

Three distinctions guide the argument. The eigenvalues of a reduced density matrix are properties of a particular state and partition; the central charge and current level are coefficients in local continuum operator algebras; and a coefficient extracted from an interval entropy or variance becomes one of those algebraic quantities only after the geometry and infrared state have been identified. In particular, “logarithmic” does not by itself mean “the homogeneous CFT vacuum.” We keep the exact Gaussian calculation separate from that interpretation, and later use a solvable defect to see why the separation matters. All logarithms are natural. Ordinary single kets denote states throughout.

A. Correlation convention and natural orbitals

Let $\mathbf{m}$ denote a complete measurement record, and $\hat{\rho}_{\mathbf{m}}$ its normalized conditioned Gaussian state. For complex fermions obeying $\{\hat{\psi}_i^\dagger, \hat{\psi}_j\} = \delta_{ij}$ and $\{\hat{\psi}_i, \hat{\psi}_j\} = 0$, define

$$(G_{\mathbf{m}})_{ij} = \text{Tr}(\hat{\rho}_{\mathbf{m}} \hat{\psi}_i^\dagger \hat{\psi}_j). \quad (1)$$

Restricting i, j to a region A gives $G_{\mathbf{m}, A}$. Here A always denotes a subsystem, not an ancilla. Positivity and the canonical anticommutation relations imply $0 \leq G_{\mathbf{m}, A} \leq \mathbf{1}_A$. For example, $v^\dagger G v$ is the expectation of $(\sum_i v_i \hat{\psi}_i)^\dagger (\sum_j v_j \hat{\psi}_j)$, and the anticommutator gives the complementary upper bound.

Temporarily suppress the record label and diagonalize

$$G_A = U \text{diag}(\nu_a) U^\dagger, \quad 0 \leq \nu_a \leq 1, \quad \hat{f}_a = \sum_i U_{ia} \hat{\psi}_i. \quad (2)$$

Then $\langle \hat{f}_a^\dagger \hat{f}_b \rangle = (U^\dagger G_A U)_{ab} = \nu_a \delta_{ab}$. The placement of conjugates in this formula is important. With the definition in Eq. (1), the occupation matrix used in the usual one-body Hamiltonian ordering is $G_A^\top$.

Partial trace preserves Gaussianity. In the natural-orbital basis the reduced state is a product of independent occupied/empty modes [2]:

$$\hat{\rho}_A = \bigotimes_a [(1 - \nu_a) |0_a\rangle \langle 0_a| + \nu_a |1_a\rangle \langle 1_a|], \quad (3)$$

$$= Z_A^{-1} \exp \left[- \sum_a \epsilon_a \hat{f}_a^\dagger \hat{f}_a \right], \quad \epsilon_a = \log \frac{1 - \nu_a}{\nu_a}, \quad Z_A = \prod_a (1 + e^{-\epsilon_a}). \quad (4)$$

One way to see the factorization is to use Wick’s theorem: in a basis with diagonal two-point functions, all occupation moments factor into products of ν_a . Gaussianity excludes independent higher connected correlations, so these moments

fix the full product density matrix. Diagonalizing a two-point function alone would not justify this conclusion for a non-Gaussian state.

In the original site basis write $\hat{\rho}_A \propto \exp[-\sum_{ij}(h_A)_{ij}\hat{\psi}_i^\dagger\hat{\psi}_j]$. Substitution of Eq. (2) gives

$$h_A = U^* \text{diag}(\epsilon_a) U^\top = \{\log[(\mathbf{1}_A - G_A)G_A^{-1}]\}^\top, \quad G_A = (\mathbf{1}_A + e^{h_A^\top})^{-1}. \quad (5)$$

Thus the transpose matters for eigenvectors and real-space modular evolution, although it does not change any spectral entropy below. Eigenvalues $\nu_a = 0, 1$ are understood by continuity: the corresponding mode is frozen and ϵ_a is infinite, not a finite matrix logarithm.

It is useful to distinguish the single-particle entanglement energies ϵ_a from the many-body modular spectrum. Defining $\tilde{\mathcal{H}}_A = -\log \hat{\rho}_A$, its eigenvalues are

$$\mathcal{E}_{\{n_a\}} = \log Z_A + \sum_a \epsilon_a n_a, \quad n_a \in \{0, 1\}. \quad (6)$$

A negative ϵ_a simply means that occupation is more likely than vacancy; the probabilities $\exp(-\mathcal{E}_{\{n_a\}})$ remain between zero and one. This modular operator is generally not the physical Hamiltonian restricted to A , and its dimensionless energies do not measure a physical excitation gap. The distinction is developed in Ref. [3], Secs. 3–5, and will be important for the defect example in Sec. VIII.

B. Taking the trace of a power

The many-body eigenvalues of Eq. (3) are $\prod_a \nu_a^{n_a} (1 - \nu_a)^{1-n_a}$ with $n_a = 0, 1$. Consequently,

$$\text{Tr } \hat{\rho}_A^q = \prod_a [\nu_a^q + (1 - \nu_a)^q], \quad (7)$$

$$S_q(A) = \frac{1}{1-q} \sum_a \log[\nu_a^q + (1 - \nu_a)^q], \quad q > 0, \quad q \neq 1. \quad (8)$$

Here q is always the Rényi index, not electric charge or a band label. Differentiating the logarithm at $q = 1$ gives

$$S_1(A) = \sum_a h_1(\nu_a), \quad h_1(\nu) = -\nu \log \nu - (1 - \nu) \log(1 - \nu), \quad (9)$$

with $0 \log 0 = 0$. For a single mode at $\nu = 1/2$, all Rényi entropies equal $\log 2$; a frozen mode contributes zero. These are exact finite-system identities, independent of criticality.

For a pure global state, the reduced-state eigenvalues are the squared Schmidt coefficients. The simplest example is the one-particle state

$$|\Psi_\nu\rangle = \sqrt{\nu}|1_A 0_{\bar{A}}\rangle + \sqrt{1-\nu}|0_A 1_{\bar{A}}\rangle. \quad (10)$$

Tracing out $\bar{A}$ gives a single factor of Eq. (3). The same uncertain location of the particle produces both $h_1(\nu)$ entanglement and $\nu(1-\nu)$ charge variance. Many Gaussian modes repeat this construction in suitable bases, but their entropy and variance add with different weights.

Increasing q emphasizes the largest many-body eigenvalues p_α . In particular,

$$S_\infty = -\log p_{\max} = -\sum_a \log \max(\nu_a, 1 - \nu_a) = \sum_a \log(1 + e^{-|\epsilon_a|}). \quad (11)$$

By contrast $S_0 = \log \text{rank } \hat{\rho}_A$ counts every nonzero eigenvalue equally. In a finite Gaussian system it is $\log 2$ times the number of modes with $0 < \nu_a < 1$. Its limit need not commute with a continuum or large-size limit: an arbitrarily small but nonzero Schmidt weight still contributes to S_0 .

If the global conditioned state is pure, these reduced-state entropies are bipartite entanglement entropies. If it is mixed, the same spectral formulas are correct, but include global mixedness. For example, $G_A = \mathbf{1}_A/2$ gives $S_q = |A| \log 2$ even for an entirely uncorrelated maximally mixed state. A logarithmic entanglement interpretation therefore needs a separate physical argument about the state, not just a Gaussian formula.

II. CHARGE STATISTICS AND MEASUREMENT-RECORD AVERAGES

A. Generating function and cumulants

In units where a physical fermion has charge one, let $\hat{Q}_A = \sum_{i \in A} \hat{\psi}_i^\dagger \hat{\psi}_i = \sum_a \hat{f}_a^\dagger \hat{f}_a$. Each natural orbital contributes a Bernoulli occupation. Its characteristic function is therefore

$$\mathcal{Z}_A(\lambda) = \text{Tr}(\hat{\rho}_A e^{i\lambda \hat{Q}_A}) = \prod_a (1 - \nu_a + e^{i\lambda} \nu_a) \quad (12)$$

$$= \det[\mathbf{1}_A - G_A + e^{i\lambda} G_A]. \quad (13)$$

This is the Gaussian full-counting-statistics determinant; Klich's trace identity supplies a general operator derivation [4]. Our product-state derivation also shows why a single-particle determinant can encode a many-body charge distribution.

Define the cumulants by $\kappa_j = (-i\partial_\lambda)^j \log \mathcal{Z}_A|_{\lambda=0}$. For one orbital, put $t = i\lambda$ and $g(t) = \log(1 - \nu + \nu e^t)$. If $p(t) = \nu e^t / (1 - \nu + \nu e^t)$, then $g' = p$ and $p' = p(1 - p)$. Repeated differentiation at $t = 0$ gives

$$\begin{aligned} \kappa_1 &= \text{tr} G_A, & F_A &\equiv \kappa_2 = \text{tr}[G_A(\mathbf{1}_A - G_A)], \\ \kappa_3 &= \text{tr}[G_A(\mathbf{1}_A - G_A)(\mathbf{1}_A - 2G_A)], & \kappa_4 &= \text{tr}[G_A(\mathbf{1}_A - G_A)(\mathbf{1}_A - 6G_A + 6G_A^2)]. \end{aligned} \quad (14)$$

In particular $F_A = \sum_a \nu_a(1 - \nu_a)$ is the quantum variance of charge in this conditioned state. Relations between free-fermion entanglement and charge statistics are developed in Refs. [5–7].

B. Which average is being taken?

Restore the record label and let an overline denote the Born average: $\overline{X} = \sum_{\mathbf{m}} p_{\mathbf{m}} X_{\mathbf{m}}$. The mean conditional entropy, annealed replica moment, and entropy of the Born mixture are three different quantities:

$$\overline{S_q(A)} = \frac{1}{1-q} \log \text{Tr} \overline{\hat{\rho}_{\mathbf{m},A}^q}, \quad (15)$$

$$S_q^{\text{ann}}(A) = \frac{1}{1-q} \log \text{Tr} \overline{\hat{\rho}_{\mathbf{m},A}^q}, \quad (16)$$

$$S_q^{\text{mix}}(A) = \frac{1}{1-q} \log \text{Tr} (\overline{\hat{\rho}_{\mathbf{m},A}})^q. \quad (17)$$

For $q > 1$, Jensen's inequality gives $\overline{S_q} \geq S_q^{\text{ann}}$; the mixture introduces cross terms between different states and is a distinct construction. For example, an equal ensemble of empty and occupied one-mode pure states has zero conditional entropy but mixture entropy $\log 2$.

A mixture of Gaussian states is generally non-Gaussian. Thus inserting $\overline{G}$ into Eq. (8) computes the entropy of the Gaussian state with that covariance, not generally that of the actual mixture. An equal mixture of the two-mode states $|00\rangle$ and $|11\rangle$ has $\overline{G} = \mathbf{1}/2$: the actual entropy is $\log 2$, whereas the Gaussian formula gives $2 \log 2$.

Charge fluctuations separate just as explicitly. Write $\mu_{\mathbf{m}} = \text{tr} G_{\mathbf{m},A}$ and $F_{\mathbf{m},A} = \langle \hat{Q}_A^2 \rangle_{\mathbf{m}} - \mu_{\mathbf{m}}^2$. The law of total variance gives

$$\text{Var}_{\text{mix}}(Q_A) = \overline{F_A} + \underbrace{\overline{\mu_{\mathbf{m}}^2} - \overline{\mu_{\mathbf{m}}}^2}_{\text{record-to-record charge wandering}}. \quad (18)$$

The current-algebra comparison below uses the first term. Averaging translations of a strip within one trajectory is another operation; when needed we denote it explicitly by $N_y^{-1} \sum_{y_0}$, reserving the overline for records.

III. CONTINUUM FERMIONS, BOSONIZATION, AND LUTTINGER LIQUIDS

Before extracting continuum data from an entropy, it is useful to specify the theories that could carry those data. We use y for the coordinate along an interface and set $\hbar = 1$. Continuum fields and their mode operators are written without hats in the standard field-theory notation; $\hat{H}$ still denotes a many-body Hamiltonian. Path-integral fermion fields are Grassmann variables. None of the Hamiltonians in this section is being identified with the modular Hamiltonian $\hat{\mathcal{H}}_A$ or with a generator of the adaptive circuit.

A. A Dirac fermion and a single chiral fermion

Linearizing a one-dimensional spinless band about its two Fermi points gives $c(y) \simeq e^{ik_F y} \psi_+(y) + e^{-ik_F y} \psi_-(y)$. The slowly varying fields obey $\{\psi_\sigma(y), \psi_\sigma^\dagger(y')\} = \delta_{\sigma\sigma'} \delta(y - y')$. We use the Weyl basis with $\psi_+ = \psi_R$, $\psi_- = \psi_L$, and $\Psi = (\psi_+, \psi_-)^T$. For equal positive velocities v_F , the normal-ordered Hamiltonian is

$$\hat{H}_D = -iv_F \int dy \left(: \psi_+^\dagger \partial_y \psi_+ : - : \psi_-^\dagger \partial_y \psi_- : \right) = \int dy : \Psi^\dagger (-iv_F \sigma^3 \partial_y) \Psi :, \quad (19)$$

$$\mathcal{S}_D = \int dt dy \left[i \psi_+^\dagger (\partial_t + v_F \partial_y) \psi_+ + i \psi_-^\dagger (\partial_t - v_F \partial_y) \psi_- \right], \quad (20)$$

$$\mathcal{S}_{D,E} = \int d\tau dy \left[\psi_+^\dagger (\partial_\tau - iv_F \partial_y) \psi_+ + \psi_-^\dagger (\partial_\tau + iv_F \partial_y) \psi_- \right]. \quad (21)$$

Normal ordering subtracts the filled reference sea; it is essential for the continuum density fluctuations. The equations of motion give $\psi_+(y, t) = \psi_+(y - v_F t, 0)$ and $\psi_-(y, t) = \psi_-(y + v_F t, 0)$. Thus the signs in the Hamiltonian specify actual propagation, not merely names for two fields [8], Sec. 3.1, Eq. (19).

A purely right-moving complex fermion retains only the $+$ term:

$$\hat{H}_R = -iv_R \int dy : \psi_R^\dagger \partial_y \psi_R :, \quad \mathcal{S}_R = \int dt dy i \psi_R^\dagger (\partial_t + v_R \partial_y) \psi_R. \quad (22)$$

Its Euclidean action is the corresponding single term in Eq. (21). A single complex chiral fermion has $(c_R, c_L) = (1, 0)$; the Dirac theory has $(1, 1)$. The customary statement “a Dirac fermion has $c = 1$ ” uses the nonchiral convention $c_R = c_L = c$, not $c_R + c_L = 1$. Likewise the physical charge-one current has level one in each occupied chiral sector. The algebraic meanings of c and k are derived below.

An isolated chiral theory is an effective edge theory with a microscopic regulator supplied, for example, by a two-dimensional bulk. It is not a standalone one-dimensional lattice band obtained by forgetting its opposite Fermi point. On a strip the opposite edge can supply the counterpropagating branch, without putting both branches at the same spatial interface. This distinction is precisely why local wall chirality can coexist with zero net chirality for the two-wall system.

B. Bosonization conventions and the fermion dictionary

We fix the bosonization normalization explicitly before using it. For a real scalar Φ with canonical momentum Π , introduce its dual φ and chiral components by

$$[\Phi(y), \Pi(y')] = i\delta(y - y'), \quad \partial_y \varphi = \Pi, \quad \Phi_\pm = \frac{\Phi \mp \varphi}{2}, \quad \Phi = \Phi_+ + \Phi_-. \quad (23)$$

On a circle the dual includes winding and a zero-mode convention; the indefinite integral of Π specifies its local derivative, not all global sectors. On the line our chiral commutators are

$$[\Phi_\pm(y), \Phi_\pm(y')] = \pm \frac{i}{4} \text{sgn}(y - y'), \quad [\Phi_+(y), \Phi_-(y')] = \frac{i}{4}. \quad (24)$$

The constant cross-commutator fixes the relative fermionic sign. With short-distance cutoff a , the dictionary is

$$\psi_\pm(y) = \frac{1}{\sqrt{2\pi a}} : e^{\pm i\sqrt{4\pi}\Phi_\pm(y)} :, \quad \rho_\pm(y) = : \psi_\pm^\dagger \psi_\pm : = \frac{1}{\sqrt{\pi}} \partial_y \Phi_\pm(y). \quad (25)$$

Here a denotes the ultraviolet length cutoff, reserving ϵ_a for modular energies; it has the role of the cutoff ϵ in the bosonization cheat sheet. For example, the commutator of the two vertex exponents is $i\pi$, so exchanging them supplies a minus sign. An alternative convention takes commuting cross-chiral bosons and introduces explicit Klein factors to supply this sign. We do not add those factors on top of Eq. (24); that would count the same sign twice. Point splitting and normal ordering give the density identity in Eq. (25); see Ref. [8], Secs. 4–5, for the operator construction.

At the free point these relations give

$$\hat{H}_B = \frac{v_F}{2} \int dy [\Pi^2 + (\partial_y \Phi)^2] = v_F \int dy [(\partial_y \Phi_+)^2 + (\partial_y \Phi_-)^2], \quad (26)$$

$$\mathcal{S}_B = \frac{1}{2} \int dt dy [v_F^{-1} (\partial_t \Phi)^2 - v_F (\partial_y \Phi)^2], \quad (27)$$

$$\mathcal{S}_{B,E} = \frac{1}{2} \int d\tau dy [v_F^{-1} (\partial_\tau \Phi)^2 + v_F (\partial_y \Phi)^2]. \quad (28)$$

One obtains the action by integrating Π out of $\int dt dy [\Pi \partial_t \Phi - \mathcal{H}]$. The conserved physical density and current are

$$\rho = \rho_+ + \rho_- = \frac{\partial_y \Phi}{\sqrt{\pi}}, \quad j = v_F (\rho_+ - \rho_-) = -\frac{v_F \Pi}{\sqrt{\pi}}, \quad \partial_t \rho + \partial_y j = 0. \quad (29)$$

The plus sign in the density and the factors of $\sqrt{\pi}$ fix our convention throughout; changing either would change the charge dictionary.

The regulated equal-time contractions are

$$G_\pm(y) = \frac{1}{4\pi} \log \frac{a}{a \mp iy}, \quad G(y) = \frac{1}{4\pi} \log \frac{a^2}{a^2 + y^2}. \quad (30)$$

These expressions fix an additive constant, equivalently a normal-ordering scale; physical neutral vertex correlators do not depend on the otherwise infrared-divergent scalar zero mode. Gaussian contraction gives $:e^A : e^B := :e^{A+B} : e^{\langle AB \rangle}$ for linear bosonic exponents in a common radial ordering. Hence a chiral vertex $:e^{i\sqrt{4\pi}\alpha\Phi} :$ has weight $h = \alpha^2/2$, while the full vertex $:e^{i\sqrt{4\pi}\alpha\Phi} :$ has $h = \bar{h} = \alpha^2/2$ and scaling dimension $\Delta = \alpha^2$. The fermion vertex at $\alpha = 1$ has chiral weight $1/2$, as required. We define these conformal terms more formally in Sec. IV.

C. The ordinary Luttinger liquid

Bosonization becomes particularly useful because density interactions, quartic in fermions, are quadratic in the boson. Consider the gapless density terms, with interbranch coupling g_B and intrabranh coupling g_F :

$$\hat{H}_{\text{int}} = \int dy [2g_B : \rho_+ \rho_- : + g_F : (\rho_+^2 + \rho_-^2) :]. \quad (31)$$

The density dictionary yields $2\rho_+ \rho_- = [(\partial_y \Phi)^2 - \Pi^2]/(2\pi)$ and $\rho_+^2 + \rho_-^2 = [(\partial_y \Phi)^2 + \Pi^2]/(2\pi)$. Adding Eq. (26) therefore gives

$$\hat{H}_{\text{LL}} = \frac{u}{2} \int dy [K_{\text{LL}} \Pi^2 + K_{\text{LL}}^{-1} (\partial_y \Phi)^2], \quad (32)$$

$$u K_{\text{LL}} = v_F + \frac{g_F - g_B}{\pi}, \quad \frac{u}{K_{\text{LL}}} = v_F + \frac{g_F + g_B}{\pi}, \quad (33)$$

$$\begin{aligned} \mathcal{S}_{\text{LL}} &= \frac{1}{2K_{\text{LL}}} \int dt dy [u^{-1} (\partial_t \Phi)^2 - u (\partial_y \Phi)^2], \\ \mathcal{S}_{\text{LL},E} &= \frac{1}{2K_{\text{LL}}} \int d\tau dy [u^{-1} (\partial_\tau \Phi)^2 + u (\partial_y \Phi)^2]. \end{aligned} \quad (34)$$

Both stiffnesses must be positive. $K_{\text{LL}} = 1$ at the noninteracting point; the density remains $\partial_y \Phi / \sqrt{\pi}$, while continuity now fixes $j = -u K_{\text{LL}} \Pi / \sqrt{\pi}$. The field rescaling $\tilde{\Phi} = \Phi / \sqrt{K_{\text{LL}}}$, $\tilde{\varphi} = \sqrt{K_{\text{LL}}} \varphi$ restores the free scalar action but cannot remove K_{LL} from the physical charge. This is the standard spinless Luttinger liquid [8], Sec. 6.1, Eqs. (115)–(119).

The propagating eigenfields are $\tilde{\Phi}_\pm = (\tilde{\Phi} \mp \tilde{\varphi})/2$. Their normalized oscillator algebras have the free form, but the physical density is $\sqrt{K_{\text{LL}}/\pi} \partial_y (\tilde{\Phi}_+ + \tilde{\Phi}_-)$. Consequently the physical current levels are $k_R = k_L = K_{\text{LL}}$, whereas $c_R = c_L = 1$. There is also a direct derivation of the variance. For an interval,

$$Q_A - \langle Q_A \rangle = \frac{\delta\Phi(\ell) - \delta\Phi(0)}{\sqrt{\pi}}, \quad F_A = \frac{K_{\text{LL}}}{\pi^2} \log \frac{D_L(\ell)}{a} + \dots, \quad D_L(\ell) = \frac{L}{\pi} \sin \frac{\pi\ell}{L}. \quad (35)$$

Here $\delta\Phi = \Phi - \langle\Phi\rangle$. The action multiplies the scalar covariance in Eq. (30) by K_{LL} ; taking the variance of the field difference and mapping to a circle gives the last expression. The entropy retains the nonchiral $c = 1$ coefficient. Thus an ordinary Luttinger liquid supplies a concrete, nonchiral theory whose charge-fluctuation coefficient varies continuously while its central charge does not [7].

Not every fermion interaction is included in this gapless quadratic action. A right-left mixing bilinear becomes a vertex with the form $\cos(\sqrt{4\pi}\Phi + \delta)$, and a commensurate spinless umklapp term has the form $\cos(\sqrt{16\pi}\Phi + \delta')$, with scaling dimensions K_{LL} and $4K_{LL}$, respectively. The phases depend on the relative fermion convention; they do not change the dimensions. The spinless four-fermion term requires point splitting, not a product of identical fields at exactly the same point. Such perturbations may gap the liquid when allowed and relevant. A single isolated chiral branch has no opposite mover with which to form this ordinary Dirac mass. For distant opposite walls, a mixing term instead requires interwall coupling.

D. The chiral Luttinger liquid and its charge normalization

The chiral theory has only one propagating density field. In the same normalization as Φ_+ , a right-moving compact boson can be written

$$\mathcal{S}_{R,M} = -M \int dt dy [\partial_t \Phi_R \partial_y \Phi_R + v_R (\partial_y \Phi_R)^2], \quad (36)$$

$$\hat{H}_{R,M} = v_R M \int dy (\partial_y \Phi_R)^2, \quad \mathcal{S}_{R,M,E} = M \int d\tau dy [i \partial_\tau \Phi_R \partial_y \Phi_R + v_R (\partial_y \Phi_R)^2]. \quad (37)$$

Here $M > 0$ is a stiffness together with a specified compact charge lattice, not the Luttinger parameter K_{LL} . The negative sign of the real-time kinetic term is fixed by our choice that a right mover satisfies $\partial_t \Phi_R = -v_R \partial_y \Phi_R$, up to a removable spatially uniform mode. Reversing spatial orientation reverses the kinetic sign, not the positive Hamiltonian. Quantizing the first-order action gives

$$[\Phi_R(y), \Phi_R(y')] = \frac{i}{4M} \text{sgn}(y - y'), \quad \rho_R = \frac{\partial_y \Phi_R}{\sqrt{\pi}}, \quad [\rho_R(y), \rho_R(y')] = -\frac{i}{2\pi M} \partial_y \delta(y - y'). \quad (38)$$

Fourier modes of this density obey the current algebra with $k_R = 1/M$, while the oscillator sector has $c_R = 1$. At $M = 1$ this is precisely the bosonized charge-one chiral complex fermion, not an additional interacting sector.

For comparison, a Laughlin edge at filling $1/M$ with odd integer M has an electron annihilator $:e^{iM\sqrt{4\pi}\Phi_R}: of weight $h = M/2$ and charge -1 . Its fundamental smaller vertex has fractional charge. This illustrates why the scalar normalization, compactification, and physical electron operator must be specified together. It is an anomalous edge supported by a topological bulk, not a standalone nonchiral compact-boson theory [9], Sec. 3.1, and [10], Secs. 6.1.3–6.1.4, with a sign reversal relative to conventions that orient the edge oppositely. In the one-component topological K -matrix notation this edge has $K_{\text{edge}} = (M)$: its quantized compactification data are not the tunable K_{LL} . For a fixed single Laughlin branch, density interactions can renormalize v_R without changing M . We are not assuming such fractional topological order for the adaptive free-fermion protocol.$

Both Luttinger descriptions are relevant to interpreting the observables, but in different ways. The ordinary liquid is the natural counterexample to equating a variance coefficient with chirality or with central charge. The $M = 1$ chiral liquid is a useful candidate low-energy description of an individual charge-one free-fermion wall. Neither bosonization nor Gaussianity proves that a conditioned circuit state is the vacuum of that local continuum theory. An interacting Luttinger liquid is Gaussian in bosons, but generally not Gaussian in the original fermions; the exact correlation-matrix entropy formulas of Sec. I cannot be extended to it merely because its bosonic action is quadratic.

IV. REPLICA DERIVATION OF THE INTERVAL ENTROPY

The Gaussian identities do not require a continuum theory. We now assume that the long-distance interval state is described by the vacuum of a $1 + 1d$ CFT on a circle, or has the same leading interval scaling. A generic monitored Gaussian state need not satisfy this assumption. The replica method and its conformal implementation are reviewed in Refs. [11, 12]. To retain chirality explicitly we follow the left/right separation discussed in Ref. [13], Sec. II.

A. The conformal ingredients

We first describe the language needed for the replica calculation. In two Euclidean dimensions a local conformal change of coordinate is an analytic map $z \mapsto f(z)$ together with its antiholomorphic counterpart. A primary field of weights $(h, \bar{h})$ obeys

$$\mathcal{O}(z, \bar{z}) = f'(z)^h \bar{f}'(\bar{z})^{\bar{h}} \mathcal{O}(f(z), \bar{f}(\bar{z})). \quad (39)$$

Its scaling dimension is $\Delta = h + \bar{h}$, while $h - \bar{h}$ is its conformal spin. Translation, scaling, and special conformal invariance then fix a conjugate pair's plane two-point function to $C_{\mathcal{O}}(z_1 - z_2)^{-2h}(\bar{z}_1 - \bar{z}_2)^{-2\bar{h}}$. The holomorphic and antiholomorphic dependencies continue to opposite propagation directions in real time. Which is called right-moving is a coordinate convention; neither label refers to the spatial position of a domain wall.

The operator product expansion organizes what happens when two insertions approach each other. The stress tensor generates local coordinate changes, and its singular product with a primary is

$$T(z)\mathcal{O}(u, \bar{u}) \sim \frac{h\mathcal{O}(u, \bar{u})}{(z-u)^2} + \frac{\partial_u \mathcal{O}(u, \bar{u})}{z-u}. \quad (40)$$

Integrating this expression around u produces the infinitesimal transformation of $\mathcal{O}$; applying the same contour to many insertions gives the Ward identity used below. This is why a singular local product can determine a global two-point scaling law. Radial quantization treats concentric circles as equal-time slices and expands

$$T(z) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}, \quad [L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}\hat{\mathbf{1}}. \quad (41)$$

The L_n generate conformal transformations; the central term commutes with all of them. There is a second commuting algebra for $\bar{T}$. The central charge is consequently not a charge eigenvalue of the state. It fixes the stress-tensor fluctuation strength and the anomalous term in its transformation law. Standard derivations of these ingredients are given in Ref. [14], Secs. 2–3, and Ref. [15].

B. From gluing replicas to twist fields

For an integer $q > 1$, the Euclidean path integral for $\hat{\rho}_A$ is cut along an interval $A = [u, v]$ at a fixed time. Multiplication of density matrices glues the upper bank of one cut to the lower bank of the next; the trace closes the sequence after q sheets. Thus

$$\text{Tr } \hat{\rho}_A^q = \frac{Z[\mathcal{R}_q]}{Z[\mathcal{R}_1]^q}, \quad (42)$$

where $\mathcal{R}_q$ is the branched surface. Equivalently, work on one plane with q copies of the CFT and insert a twist field $\mathcal{T}_q(u)$ and an inverse twist $\tilde{\mathcal{T}}_q(v)$. Encircling one insertion cycles the copy index. The ratio in Eq. (42) is their two-point function, with cutoff-dependent normalization.

To find the twist dimension for one holomorphic branch of central charge c , uniformize the branched surface with

$$f(z) = \left(\frac{z-u}{z-v} \right)^{1/q}. \quad (43)$$

A loop around u changes sheets, while f resolves the branch into a single-valued local coordinate. The holomorphic stress tensor transforms as

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12} \{f, z\}, \quad \{f, z\} = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2. \quad (44)$$

The plane vacuum has $\langle T(f) \rangle = 0$. Since a fractional linear map has zero Schwarzian and $\{g^\alpha, g\} = (1 - \alpha^2)/(2g^2)$, Eq. (43) gives

$$\{f, z\} = \frac{1 - q^{-2}}{2} \frac{(u-v)^2}{(z-u)^2(z-v)^2}. \quad (45)$$

The replica stress tensor sums over all q sheets. Therefore

$$\langle T_{\text{tot}}(z) \rangle_{\mathcal{R}_q} = \frac{c}{24} \left(q - \frac{1}{q} \right) \frac{(u-v)^2}{(z-u)^2(z-v)^2}. \quad (46)$$

For two primary insertions of weight h , the conformal Ward identity expresses a stress-tensor insertion in terms of the two-point function:

$$\begin{aligned} \frac{\langle T_{\text{tot}}(z) \mathcal{T}(u) \tilde{\mathcal{T}}(v) \rangle}{\langle \mathcal{T}(u) \tilde{\mathcal{T}}(v) \rangle} &= \sum_{\zeta=u,v} \left[\frac{h}{(z-\zeta)^2} + \frac{1}{z-\zeta} \partial_{\zeta} \log(u-v)^{-2h} \right] \\ &= h \frac{(u-v)^2}{(z-u)^2(z-v)^2}. \end{aligned} \quad (47)$$

The derivatives contribute $-2h/(u-v)$ and $2h/(u-v)$, respectively; combining the fractions gives the last line. Matching Eq. (46) then determines

$$h_q = \frac{c}{24} \left(q - \frac{1}{q} \right). \quad (48)$$

The factor of q from the replica stress tensor is essential. Missing it would give the wrong Rényi coefficient. If both chiralities are present, their weights are obtained separately with c_R and c_L ; their sum sets the equal-time decay exponent.

C. Plane-to-cylinder map and the Rényi coefficient

On the plane, a single chiral twist pair has correlator proportional to $(z_1 - z_2)^{-2h_q}$. Let the circle circumference be $L = N_y$ in lattice units, take $w = \tau + iy$ with Euclidean time rescaled by the branch velocity, and map

$$z = \exp(2\pi w/L), \quad \mathcal{T}_q^{\text{cyl}}(w) = \left(\frac{dz}{dw} \right)^{h_q} \mathcal{T}_q(z). \quad (49)$$

For equal-time endpoints at $y = 0, \ell$, $|z_1 - z_2| = 2 \sin(\pi \ell/L)$ and $|z'_1 z'_2|^{h_q} = (2\pi/L)^{2h_q}$. Their ratio gives

$$\text{Tr } \hat{\rho}_{[0,\ell]}^q = C_q \left[\frac{L}{\pi a} \sin \left(\frac{\pi \ell}{L} \right) \right]^{-2h_q}, \quad X_L(\ell) = \log \left[\frac{L}{\pi a} \sin \left(\frac{\pi \ell}{L} \right) \right]. \quad (50)$$

The microscopic cutoff a and C_q encode endpoint physics; the trace uses a consistent regulator supplied by the microscopic system. Taking $\log/(1-q)$ and using $(q-1/q)/(q-1) = 1 + 1/q$ gives, for a single branch,

$$S_q^{(w)}(\ell) = \frac{c_w}{12} \left(1 + \frac{1}{q} \right) X_L(\ell) + b_{w,q} + \dots. \quad (51)$$

Analytic continuation from integer replicas to $q > 0$ is the usual replica assumption. In the limit $q \rightarrow 1$ the coefficient is $c_w/6$.

The geometry is part of this result. There are two entangling endpoints in an interval on a circle, even for one chiral branch. A nonchiral system on a half-line, with an interval ending on a physical boundary, instead uses a twist one-point function in the half-plane. Its logarithmic coefficient is half that of a bulk interval, with an additional boundary constant [12], Sec. 3.2. A defect placed at an entangling endpoint is a further change: the homogeneous two-point function above no longer suffices. The half-chain defect calculation below must therefore be compared with its own homogeneous half-chain, not with the full-strip coefficient by a bare numerical identification.

For independent branches intersected by the region, leading contributions add. Defining c_{Σ} as the sum over both orientations and all walls,

$$S_q(\ell) = a_{S_q} X_L(\ell) + b_q + \dots, \quad a_{S_q} = \frac{c_{\Sigma}}{12} \left(1 + \frac{1}{q} \right). \quad (52)$$

Independence is an infrared assumption: interwall coupling or extra gapless sectors changes the content being counted. Finite temperature, an excited state, or a finite correlation length can also replace the vacuum chord law. Gaussianity alone fixes none of these features.

In a monitored ensemble, the same coefficient passes to $\overline{S}_q$ if the contributing records share it and their averaged corrections remain subleading. A logarithmic ensemble mean alone does not prove the same conformal description for every record. This use of Eq. (52) always concerns the conditional average of Eq. (15), not the entropy of the mixture.

V. THE PHYSICAL CURRENT AND ITS KAC-MOODY LEVEL

A. From a symmetry to the full affine algebra

A conserved internal current generates rotations of fields, whereas the stress tensor generates changes of coordinates. Their algebras therefore contain different information. Begin with generators t^a of a compact Lie algebra $\mathfrak{g}$, with $[t^a, t^b] = if^{ab}_c t^c$ and invariant inner product κ^{ab} . The indices label symmetry generators, not lattice orbitals. The holomorphic currents have the singular product

$$J^a(z)J^b(u) \sim \frac{k\kappa^{ab}}{(z-u)^2} + \frac{if^{ab}_c J^c(u)}{z-u}, \quad J_n^a = \oint \frac{dz}{2\pi i} z^n J^a(z). \quad (53)$$

The simple pole reproduces the original Lie algebra; the double pole encodes its central extension. Here and below CFT fields and modes use the standard unhatted notation, although they are quantum operators; their central identity is still $\hat{\mathbf{1}}$.

To see explicitly how the local product gives a commutator, subtract the two radial orderings of the contours. The remaining inner contour surrounds u and selects the singular part:

$$\begin{aligned} [J_m^a, J_n^b] &= \oint \frac{du}{2\pi i} u^n \oint_u \frac{dz}{2\pi i} z^m \left[\frac{k\kappa^{ab}\hat{\mathbf{1}}}{(z-u)^2} + \frac{if^{ab}_c J^c(u)}{z-u} \right] \\ &= if^{ab}_c J_{m+n}^c + km\kappa^{ab}\delta_{m+n,0}\hat{\mathbf{1}}. \end{aligned} \quad (54)$$

The double pole differentiates z^m ; the final contour is nonzero only when $m+n=0$. In particular, the zero modes satisfy the ordinary global symmetry algebra with no central term. The nonzero modes describe its spatially varying current fluctuations. This is the affine Kac-Moody algebra in a representation of level k [14], Sec. 9.1, Eqs. (9.1)–(9.2).

For completeness, before selecting a representation the central number is replaced by a generator $\mathbf{K}$. The extended loop algebra also includes a grading derivation d :

$$\begin{aligned} [J_m^a, J_n^b] &= if^{ab}_c J_{m+n}^c + m\kappa^{ab}\delta_{m+n,0}\mathbf{K}, \\ [\mathbf{K}, J_n^a] &= 0, \quad [d, \mathbf{K}] = 0, \quad [d, J_n^a] = nJ_n^a. \end{aligned} \quad (55)$$

In a fixed-level irreducible representation $\mathbf{K} = k\hat{\mathbf{1}}$. The derivation records the mode number; it is not an additional physical charge. Since $[L_0, J_n^a] = -nJ_n^a$, it may be represented by $-L_0$ up to a commuting constant. These definitions distinguish the abstract affine algebra from its realization in a particular CFT [16], Definitions 3.3 and 3.5 and Sec. 4.2.

An antiholomorphic sector has a separate copy of Eq. (54), and the two copies commute. Using R, L for physical propagation directions, their levels are k_R, k_L ; conformal symmetry alone does not require them to be equal. Each current is a weight-one primary, so the combined conformal and current algebra also contains

$$[L_m, J_n^a] = -nJ_{m+n}^a, \quad [L_m, \bar{J}_n^a] = [J_m^a, \bar{J}_n^b] = 0, \quad (56)$$

with the corresponding barred relation. Together with the two Virasoro algebras, this specifies the symmetry structure used here. The first commutator follows from the weight-one version of Eq. (40) [14], Eq. (9.8).

B. The Abelian current, its states, and physical normalization

For $U(1)$ there is one generator and $f^{ab}_c = 0$. Set its metric to one, obtaining

$$J(z)J(0) \sim \frac{k}{z^2}, \quad J(z) = \sum_{n \in \mathbb{Z}} J_n z^{-n-1}, \quad (57)$$

and

$$[J_m, J_n] = km\delta_{m+n,0}\hat{\mathbf{1}}, \quad J_n^\dagger = J_{-n}. \quad (58)$$

Thus the Abelian algebra is a set of harmonic oscillators together with the conserved charge J_0 . A current highest-weight state is defined by

$$J_{n>0}|Q\rangle = 0, \quad J_0|Q\rangle = Q|Q\rangle. \quad (59)$$

Applying J_{-n} creates a density-wave excitation without changing Q . For a normalized highest-weight state, $\|J_{-n}|Q\rangle\|^2 = \langle Q|[J_n, J_{-n}]|Q\rangle = kn$ for $n > 0$. Unitarity requires $k > 0$ for a nontrivial current. It does not, from this oscillator calculation alone, require an integer k . The word “level” refers to the central-extension coefficient, not an occupation number or an individual energy level.

The physical charge normalization is fixed by its action on local operators. If $\mathcal{O}_Q$ carries charge Q , then

$$J(z)\mathcal{O}_Q(u) \sim \frac{Q\mathcal{O}_Q(u)}{z-u}, \quad [J_0, \mathcal{O}_Q] = Q\mathcal{O}_Q. \quad (60)$$

This contour Ward identity connects a continuum normalization to the microscopic number operator. A physical annihilation field has charge -1 and its adjoint has charge $+1$.

For a charge-one chiral complex fermion with propagator $\langle\psi(z)\psi^\dagger(0)\rangle = 1/z$, the normal-ordered bilinear current $J = :\psi^\dagger\psi:$ has double contraction $1/z^2$. Thus $k = 1$. Normal ordering removes the coincident vacuum contraction. The plane fermion here has a unit-normalized propagator; the conversion to the physical density retains a factor $1/(2\pi)$ in Eq. (61). This is compatible with the $1/\sqrt{2\pi a}$ in Eq. (25), not a change of the physical unit charge. The normalization is physical, not just notation: replacing J by ηJ changes k to $\eta^2 k$ and changes the charge generator. Throughout, the microscopic number operator fixes the unit charge. For compact simple non-Abelian symmetry, positive-energy unitary integrable representations require nonnegative integer level in the basic normalization of the invariant form [14], Sec. 9.3. This statement must not be transplanted without qualification to a physical $U(1)$ component. Compactification, the allowed charge lattice, and the combination of chiralities matter. Indeed, a nonchiral compact boson has continuously variable equal left/right physical current coefficients as its radius is varied [17], Sec. 2.3. An integer coefficient in a free-fermion example is therefore neither a general theorem about Abelian currents nor a diagnostic of chirality.

C. Density correlator on a circle

A current is a primary of holomorphic weight one, so under Eq. (49),

$$J_{\text{cyl}}(w) = \frac{2\pi}{L} z J(z), \quad \hat{\rho}_w(y) = \frac{1}{2\pi} J_{\text{cyl}}(iy). \quad (61)$$

The subscript w labels the wall branch; the argument of J_{cyl} is the Euclidean coordinate. With real-time continuation $\tau = it$, the choice $\tau + iy$ is the left-moving orientation: a holomorphic field depends on $t + y$. The right mover of Eq. (22) instead uses $\tau - iy$. All equal-time variances below are the same for either orientation; their mode-to-density Fourier signs are opposite. The factor $1/(2\pi)$ makes $\int_0^L dy \hat{\rho}_w(y) = J_0$ for the fluctuating charge relative to the reference background. At separated points,

$$\langle J_{\text{cyl}}(w) J_{\text{cyl}}(0) \rangle_c = k \left(\frac{2\pi}{L} \right)^2 \frac{z}{(z-1)^2}. \quad (62)$$

At equal time, $z = e^{2\pi iy/L}$ and $z/(z-1)^2 = -1/[4 \sin^2(\pi y/L)]$. Therefore

$$C_w(y) \equiv \langle \hat{\rho}_w(y) \hat{\rho}_w(0) \rangle_c = -\frac{k_w}{4\pi^2} \left(\frac{\pi}{L} \right)^2 \csc^2 \left(\frac{\pi y}{L} \right), \quad y \neq 0. \quad (63)$$

The antiholomorphic branch gives the same equal-time expression. Its orientation does not change the sign of a charge variance. The negative separated-point correlation is compatible with a positive variance: a positive short-distance contact contribution must also be included. Equation (63) is not an unregulated formula at $y = 0$.

D. Integrating the correlator

Let $\hat{Q}_{[0,\ell]} = \int_0^\ell dy \hat{\rho}_w(y)$, with a short-distance regulator. Translation invariance and the symmetry of the equal-time connected correlator imply

$$F_w(\ell) = \int_0^\ell dy \int_0^\ell dy' C_w(y-y'), \quad F_w''(\ell) = 2C_w(\ell) \quad (64)$$

away from the regulated endpoint region. Since $d^2 \log \sin(\pi \ell/L)/d\ell^2 = -(\pi/L)^2 \csc^2(\pi \ell/L)$, integration gives

$$F_w(\ell) = \frac{k_w}{2\pi^2} \log \sin \left(\frac{\pi \ell}{L} \right) + A_w \ell + B_w. \quad (65)$$

The separated-point correlator alone leaves the linear term undetermined. In a definite-total-charge state, $Q_{\bar{A}} = Q_{\text{tot}} - Q_A$ gives $F(\ell) = F(L - \ell)$, after invoking translation invariance. This symmetry sets $A_w = 0$ for an independent fixed-charge branch. The contact term and cutoff then enter the additive constant:

$$F_w(\ell) = \frac{k_w}{2\pi^2} X_L(\ell) + b_{F,w} + \dots, \quad a_F = \frac{k_\Sigma}{2\pi^2}. \quad (66)$$

Here $k_\Sigma = \sum_w k_w$ counts independent branches. If only the combined system has definite total charge, the argument applies to its full variance; separate wall variances require additional independence and zero-mode assumptions. Grand-canonical mixing, fluctuating zero modes, or finite-temperature states can supply other terms. Dropping the linear term is consequently a physical condition, not just a choice of integration constant.

The CFT expression is valid for $a \ll \ell, L - \ell$. It is not to be extrapolated to $\ell = 0$ or L , where the exact lattice variance is zero and the logarithmic continuum approximation is invalid.

E. A regulated mode sum: where the contact term went

The oscillator algebra supplies a complementary derivation that keeps positivity and the endpoints visible. Equation (61) gives $\hat{\rho}_w(y) = L^{-1} \sum_n J_n e^{-2\pi i n y/L}$. In a highest-weight state of fixed charge, the zero mode has no variance and $\langle J_n J_{-n} \rangle_c = kn$ for $n > 0$. Integrating the Fourier modes over $[0, \ell]$ and regulating them by $r^n = \exp(-2\pi a n/L)$ gives

$$\begin{aligned} F_{w,a}(\ell) &= \frac{k_w}{\pi^2} \sum_{n=1}^{\infty} \frac{r^n}{n} \sin^2 \frac{\pi n \ell}{L} \\ &= \frac{k_w}{4\pi^2} \log \left[1 + \frac{4r \sin^2(\pi \ell/L)}{(1-r)^2} \right]. \end{aligned} \quad (67)$$

To obtain the second line, write $2 \sin^2 \theta = 1 - \cos 2\theta$ and use $\sum_{n>0} z^n/n = -\log(1-z)$. This expression is nonnegative, symmetric under $\ell \leftrightarrow L - \ell$, and vanishes at both endpoints. For $a \ll \ell, L - \ell$, $1 - r \simeq 2\pi a/L$ reproduces Eq. (66). It is an explicit continuum regulator, not a universal finite-lattice formula. The short-distance contribution needed to compensate the negative separated-point correlator is already contained in the regulated sum. A symmetrized density covariance may be used throughout this variance calculation; the antisymmetric contact commutator integrates to zero against the same interval on both sides.

The zero-mode assumption can now be read off rather than hidden in an integration constant. An incoherent mixture of highest-weight sectors with fluctuating J_0 contributes

$$F_{w,a}^{\text{mix}}(\ell) = F_{w,a}(\ell) + \left(\frac{\ell}{L} \right)^2 \text{Var}(J_0). \quad (68)$$

Excited oscillator populations give further terms. Fixed charge alone does not make an arbitrary state a current vacuum; both conditions entered the derivation. Equation (68) is also a continuum example of the conditional-versus-mixture distinction in Eq. (18).

VI. WHY CENTRAL CHARGE AND CURRENT LEVEL ARE DISTINCT

The central charge normalizes the stress-tensor OPE,

$$T(z)T(0) \sim \frac{c/2}{z^4} + \frac{2T(0)}{z^2} + \frac{\partial T(0)}{z}. \quad (69)$$

The current level instead normalizes Eq. (57). Their relation can be understood directly for an Abelian current.

A. Sugawara construction and charged descendants

Define its Sugawara stress tensor by

$$T_J(z) = \frac{1}{2k} :J(z)J(z):. \quad (70)$$

The two possible contractions with $J(0)$ give $T_J(z)J(0) \sim J(0)/z^2 + \partial J(0)/z$, so the current has weight one under this stress tensor. In $T_J(z)T_J(0)$ there are two double contractions and four single contractions. Their coefficients are

$$\frac{2k^2}{(2k)^2 z^4} = \frac{1}{2z^4}, \quad \frac{4k}{(2k)^2 z^2} :J(z)J(0): = \frac{2T_J(0)}{z^2} + \frac{\partial T_J(0)}{z} + \dots \quad (71)$$

Comparison with Eq. (69) gives $c_J = 1$, independent of $k > 0$. This is the Abelian Sugawara construction [15], and already shows why $c = k$ is not a general identity.

The corresponding modes display what this stress tensor counts:

$$L_n^{(J)} = \frac{1}{2k} \sum_{p \in \mathbb{Z}} :J_{n-p}J_p:, \quad L_0^{(J)} = \frac{J_0^2}{2k} + \frac{1}{k} \sum_{p>0} J_{-p}J_p. \quad (72)$$

With $b_p = J_p/\sqrt{kp}$ for $p > 0$, the second term is $\sum_{p>0} p b_p^\dagger b_p$. It counts density-wave excitations above the charged highest-weight state; the first term is its charge cost. Thus the eigenvalue of $L_0^{(J)}$ on a descendant is $Q^2/(2k) + \sum_{p>0} p N_p$. The cylinder energy of this chiral sector is $(2\pi v/L)(L_0^{(J)} - 1/24)$, with the last term the Casimir shift. This makes the current representation an explicit spectrum rather than only a commutator.

For a compact simple non-Abelian algebra, now choose κ to be the basic invariant metric (long roots have squared length two), and define the level k relative to that choice. The analogous result is

$$T_{\mathfrak{g}} = \frac{\kappa_{ab} :J^a J^b:}{2(k + h^\vee)}, \quad c_{\mathfrak{g}} = \frac{k \dim \mathfrak{g}}{k + h^\vee}. \quad (73)$$

Here κ_{ab} is the inverse metric and $h^\vee$ is the dual Coxeter number. The shift $h^\vee$ is the extra contribution of the non-Abelian simple pole when normal ordering the currents. It is absent for $U(1)$, recovering Eq. (70). Specifying the metric matters: rescaling it changes the number called k . These formulas and their normal-ordering derivation are given in Ref. [14], Sec. 9.2.

B. Neutral sectors and the limits of coefficient matching

When the full theory contains a commuting neutral sector, $T = T_J + T_{\text{neutral}}$ and $c = 1 + c_{\text{neutral}}$, while the physical $U(1)$ current still has level k . Conversely, rescaling the physical charge of a single species changes k quadratically but leaves its central charge unchanged. A charged primary with $J(z)\mathcal{O}_Q(0) \sim Q\mathcal{O}_Q(0)/z$ has charged-sector weight $h_Q = Q^2/(2k)$; a neutral factor can add to its total weight. This follows either by the double contraction in Eq. (70) or by acting with its zero mode on a current-algebra highest-weight state.

For one charge-one free complex fermion, the current has $k = 1$ and its stress tensor is exhausted by T_J , giving $c = 1$. Independent copies add both coefficients. This special theory predicts

$$a_{S_q} = \frac{\pi^2}{6} \left(1 + \frac{1}{q}\right) a_F, \quad a_{S_q} - \frac{\pi^2}{6} \left(1 + \frac{1}{q}\right) a_F = \frac{c_\Sigma - k_\Sigma}{12} \left(1 + \frac{1}{q}\right). \quad (74)$$

These relations concern leading logarithmic coefficients. Equality of two coefficients by itself does not reconstruct an operator spectrum or uniquely identify a CFT; it is a consequence of the specified free-fermion content, not its converse.

VII. A COMMON SUSCEPTIBILITY SPECTRUM

A. Exact spectral relations

There is also a microscopic reason why the entropy and variance are closely related in a Gaussian state. Define

$$\mathcal{V}_A = G_A(\mathbf{1}_A - G_A), \quad v_a = \nu_a(1 - \nu_a) \in [0, 1/4]. \quad (75)$$

Because each entropy kernel is symmetric under $\nu \leftrightarrow 1 - \nu$, it is a function of v :

$$F_A = \text{tr } \mathcal{V}_A, \quad S_q(A) = \text{tr } g_q(\mathcal{V}_A), \quad g_q(v) = \frac{\log[\nu(v)^q + (1 - \nu(v))^q]}{1 - q}, \quad \nu(v) = \frac{1 + \sqrt{1 - 4v}}{2}. \quad (76)$$

The other root gives the same entropy. For example,

$$S_2 = - \sum_a \log(1 - 2v_a), \quad S_3 = - \frac{1}{2} \sum_a \log(1 - 3v_a). \quad (77)$$

Their small- v expansions begin with $2v$ and $3v/2$, respectively, whereas F weights every v linearly. Thus the full susceptibility spectrum fixes all Gaussian Rényi entropies, but its first moment F does not. The spectra $\{v_a\} = \{1/4, 0\}$ and $\{1/8, 1/8\}$ have equal variance and different S_2 . The map to v also loses whether each occupation lies above or below $1/2$, so it cannot determine the mean charge or odd cumulants.

For a globally pure Slater determinant, $G^2 = G$. Taking the AA block of this identity gives

$$\mathcal{V}_A = G_{A\bar{A}} G_{\bar{A}A}, \quad F_A = \|G_{A\bar{A}}\|_F^2. \quad (78)$$

The v_a are the squared singular values of the cross-boundary block. For a mixed Gaussian state the exact replacement is

$$\mathcal{V}_A = G_{A\bar{A}} G_{\bar{A}A} + (G - G^2)_{AA}. \quad (79)$$

The positive second term records intrinsic mixedness, making explicit why the same variance need not represent entanglement in a mixed state.

B. Modular-energy kernels and their integrals

Use $\epsilon = \log[(1 - \nu)/\nu]$ to write

$$f(\epsilon) = \nu(1 - \nu) = \frac{1}{4 \cosh^2(\epsilon/2)}, \quad (80)$$

$$s_q(\epsilon) = \frac{\log[2 \cosh(q\epsilon/2)] - q \log[2 \cosh(\epsilon/2)]}{1 - q}, \quad (81)$$

$$s_1(\epsilon) = \log[2 \cosh(\epsilon/2)] - \frac{\epsilon}{2} \tanh(\epsilon/2). \quad (82)$$

With the unnormalized spectral density $\varrho_A(\epsilon) = \sum_a \delta(\epsilon - \epsilon_a)$,

$$F_A = \int d\epsilon \varrho_A(\epsilon) f(\epsilon), \quad S_q(A) = \int d\epsilon \varrho_A(\epsilon) s_q(\epsilon). \quad (83)$$

It is the density including the number of modes, not a unit-area histogram, that enters these equations.

The variance kernel integrates to one because $f(\epsilon) = -d\nu/d\epsilon$ and ν falls from one to zero. For the entropy kernel, evenness and $\epsilon > 0$ give

$$s_q(\epsilon) = \frac{\log(1 + e^{-q\epsilon}) - q \log(1 + e^{-\epsilon})}{1 - q}. \quad (84)$$

For $b > 0$, expanding the logarithm and integrating term by term yields

$$\int_0^\infty d\epsilon \log(1 + e^{-b\epsilon}) = \frac{1}{b} \sum_{n=1}^\infty \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12b}. \quad (85)$$

It follows that

$$\int_{-\infty}^\infty f(\epsilon) d\epsilon = 1, \quad \int_{-\infty}^\infty s_q(\epsilon) d\epsilon = \frac{\pi^2}{6} \frac{q^{-1} - q}{1 - q} = \frac{\pi^2}{6} \left(1 + \frac{1}{q}\right). \quad (86)$$

The continuous $q = 1$ value is $\pi^2/3$.

Suppose the coefficient of the growing spectral density is approximately energy independent over the kernel-supporting energies, $\varrho_A(\epsilon) = \gamma X_L(\ell) + \varrho_{\text{sub}}(\epsilon, \ell)$, where the remainder contributes $o(X_L)$ to both integrals. Then Eq. (86) reproduces Eq. (74). This is an additional spectral assumption, not a theorem for every Gaussian state. More generally an energy-dependent coefficient $\gamma(\epsilon)$ is weighted differently by s_q and f . A finite density near zero or a logarithmically growing count in one energy window does not establish an energy-independent coefficient. Nor are the two kernels proportional pointwise. The derivation explains a leading-coefficient relation without asserting $S_q(\ell) \propto F(\ell)$ at every finite interval length.

C. Why a homogeneous critical chain can have a flat coefficient

The spectral assumption has a familiar controlled realization. For a long interval of a homogeneous critical hopping chain, the active single-particle modular levels are asymptotically equally spaced, with spacing of order $\pi^2/\log L$ on each side of zero. Thus the signed modular-energy density has leading value $\log L/\pi^2$ in that two-endpoint geometry [3], Sec. 4.1, Eq. (28). Inserting this value in Eq. (86) gives $F \simeq \log L/\pi^2$ and $S_q \simeq (1+q^{-1})\log L/6$, the nonchiral Dirac coefficients. The logarithm counts the number of partially occupied modes on which the kernels have appreciable weight. Neither all lattice modes nor just the mode closest to $\nu = 1/2$ determines that coefficient. A defect makes this explanation more informative by changing the energy coordinate in which the levels are approximately uniform.

VIII. A SOLVABLE INTERFACE DEFECT: LOGARITHMS WITHOUT HOMOGENEOUS COEFFICIENTS

Eisler and Peschel's solution [18] gives a concrete test of the preceding reasoning. Their principal example is a critical transverse Ising (TI) chain of $2L_h$ sites, divided into two halves at a modified bond. The number-conserving XX hopping chain is obtained from two Ising sectors and has twice the leading entropy. This is an equilibrium defect *at an entanglement interface*, not a demonstration that an adaptive domain wall or a hard-wall measurement support realizes the same scatterer.

A. The spectral map and the logarithmic mode count

Write $t_d > 0$ for the defect bond relative to the homogeneous bond, and introduce

$$s = \frac{2t_d}{1+t_d^2}, \quad 0 \leq s \leq 1, \quad \mathcal{T}_{\text{def}} = s^2. \quad (87)$$

The last equality is the Fermi-energy transmission probability: it follows from the scattering phases in Ref. [18], Eqs. (31)–(32). Strong and weak bonds t_d and $1/t_d$ have the same s . The spectral relation in their Eqs. (17)–(20) is

$$\cosh \Omega_s(\xi) = \frac{\cosh \xi}{s}, \quad \xi \geq 0, \quad |\epsilon| = 2\Omega_s(\xi). \quad (88)$$

We renamed their homogeneous half-energy ε as ξ . Their transfer matrix supplies Ω_s ; the reduced density matrix requires its square, explaining the factor of two in the actual modular energy ϵ . Missing this factor would change every entropy integral.

The geometry behind their Eq. (21) is also instructive. Mapping an annulus to a strip by a logarithm produces a strip width proportional to $\log L_h$. In the notation of that paper its aspect ratio obeys $2M/N = 2\pi/\log L_h$, and hence

$$\sum_{\text{TI modes}} \longrightarrow \frac{N}{M\pi} \int_0^\infty d\xi = \frac{\log L_h}{\pi^2} \int_0^\infty d\xi. \quad (89)$$

The measure is flat in the *reference half-energy* ξ , not in the physical defect energy $2\Omega_s(\xi)$. The logarithmic prefactor survives because the mapped strip becomes wider with size even though the dispersion in that strip has changed. Boundary quantization shifts do not affect this leading continuum density.

Using the even modular kernels from Eq. (82), define

$$\mathcal{I}_q(s) = \int_0^\infty d\xi s_q(2\Omega_s(\xi)). \quad (90)$$

Then, for this single-cut half-chain geometry,

$$S_q^{\text{TI}} = \frac{\mathcal{I}_q(s)}{\pi^2} \log L_h + O(1), \quad S_q^{\text{XX}} = \frac{2\mathcal{I}_q(s)}{\pi^2} \log L_h + O(1). \quad (91)$$

The XX factor of two already includes its two Ising sectors; one must not double it again when using a signed modular-energy density. For a generic bond defect these relations describe the leading scaling limit. The follow-up [19], Sec. II, constructs special scale-free defects for which the spectral identities hold more directly at finite size. They are not identities for arbitrary inhomogeneous Gaussian matrices.

B. The entropy integral and effective central charge

At $q = 1$, Eq. (90) is precisely the $I = I_1 + I_2$ of Ref. [18], Eqs. (22)–(23):

$$\mathcal{I}_1(s) = \int_0^\infty d\xi \left[\log(1 + e^{-2\Omega_s}) + \frac{2\Omega_s}{1 + e^{2\Omega_s}} \right]. \quad (92)$$

Their crucial simplification is to differentiate with respect to s . The two needed derivatives are

$$\partial_s \Omega_s = -\frac{\coth \Omega_s}{s}, \quad \frac{d}{d\Omega} s_1(2\Omega) = -\Omega \operatorname{sech}^2 \Omega, \quad \mathcal{I}'_1(s) = \frac{1}{s} \int_0^\infty \frac{\Omega_s d\xi}{\sinh \Omega_s \cosh \Omega_s}. \quad (93)$$

Differentiating once more and substituting $x = \cosh \Omega_s = \cosh \xi/s$ produces the explicit intermediate integral in their Eq. (24):

$$\mathcal{I}''_1(s) = \frac{1}{s^2} \int_{1/s}^\infty \frac{dx}{\sqrt{x^2 - s^{-2}}} \left[\frac{x \operatorname{arccosh} x}{(x^2 - 1)^{3/2}} - \frac{1}{x^2 - 1} \right]. \quad (94)$$

The bracket equals $-d[\operatorname{arccosh} x/\sqrt{x^2 - 1}]/dx$. Evaluating the convergent integral, with the combined endpoint terms kept together when performing integrations by parts, gives their Eq. (25):

$$\mathcal{I}''_1(s) = -\frac{\log s}{1 - s^2}, \quad \mathcal{I}_1(0) = \mathcal{I}'_1(0) = 0. \quad (95)$$

The boundary conditions fix both integration constants. Integrating twice, or differentiating the expression below to verify them, yields

$$\begin{aligned} \mathcal{I}_1(s) = & -\frac{1}{2} \{ [(1+s) \log(1+s) + (1-s) \log(1-s)] \log s \\ & + (1+s) \operatorname{Li}_2(-s) + (1-s) \operatorname{Li}_2(s) \}, \quad \operatorname{Li}_2(z) = -\int_0^z \frac{\log(1-x)}{x} dx. \end{aligned} \quad (96)$$

Limits at $s = 0, 1$ are taken continuously. In particular $\mathcal{I}_1(1) = \pi^2/12$, so the transparent TI half-chain has $S_1 = \log L_h/12$ and the transparent XX half-chain has $S_1 = \log L_h/6$, as their respective bulk central charges $1/2$ and 1 require for one entangling cut.

For the XX problem it is convenient to write

$$S_1^{\text{XX}} = \frac{c_{\text{eff}}(s)}{6} \log L_h + O(1), \quad c_{\text{eff}}(s) = \frac{12\mathcal{I}_1(s)}{\pi^2}. \quad (97)$$

This coefficient varies continuously from zero to one. The bulk remains the same critical Dirac theory: its local stress-tensor algebra still has $c_R = c_L = 1$. What changed is transmission across the entanglement interface. An effective entropy coefficient is not automatically a Virasoro central charge, a current level, or a count of chiral branches.

The Rényi dependence is equally explicit. Evaluating Eq. (90), as in Ref. [19], Sec. IV, gives

$$\begin{aligned} \mathcal{I}_2(s) &= \arcsin^2(s/\sqrt{2}), & \mathcal{I}_3(s) &= \frac{1}{2} \arcsin^2(\sqrt{3}s/2), \\ \mathcal{I}_{1/2}(s) &= \frac{1}{2} \arcsin s (\pi - \arcsin s), & \mathcal{I}_\infty(s) &= \frac{1}{4} \operatorname{Li}_2(s^2). \end{aligned} \quad (98)$$

For example, $s_2(2\Omega) = -\log[1 - s^2/(2 \cosh^2 \Omega)]$. If $A(b) = -\int_0^\infty \log(1 - b \operatorname{sech}^2 \xi) d\xi$, the substitution $u = \tanh \xi$ gives $A'(b) = \arcsin \sqrt{b}/\sqrt{b(1-b)}$. Since $A(0) = 0$, $A(b) = \arcsin^2 \sqrt{b}$, proving the $q = 2$ formula; the $q = 3$ kernel gives the next one in the same way. The $q = \infty$ kernel is Eq. (11). Its integral is the first term of the paper's $I_1 + I_2$, not our von Neumann $\mathcal{I}_1$.

At transparency $\mathcal{I}_q(1) = \pi^2(1 + q^{-1})/24$, recovering the homogeneous Rényi relation. Away from it, a single q -independent replacement of c in the homogeneous formula does not in general reproduce all Rényi indices. For example,

$$\mathcal{I}_1(s) = s^2 \left(\frac{3}{4} - \frac{1}{2} \log s \right) + O(s^4 |\log s|), \quad \mathcal{I}_q(s) \sim \frac{q}{4(q-1)} s^2 \quad (q > 1, s \rightarrow 0). \quad (99)$$

The weak-transmission and $q \rightarrow 1$ limits are nonuniform. The $q = 0$ rank entropy requires a separate order-of-limits analysis.

C. Susceptibility, charge fluctuations, and the nonflat density

The same spectral map immediately implies

$$v_{\text{def}} = \frac{1}{4 \cosh^2 \Omega_s(\xi)} = s^2 \frac{1}{4 \cosh^2 \xi} = \mathcal{T}_{\text{def}} v_{\text{ref}}. \quad (100)$$

This is the occupation-product relation derived in Ref. [19], Sec. II, Eqs. (6)–(9). For the number-conserving XX chain it gives

$$F^{\text{XX}} = \frac{2 \log L_h}{\pi^2} \int_0^\infty \frac{s^2 d\xi}{4 \cosh^2 \xi} + O(1) = \frac{\mathcal{T}_{\text{def}}}{2\pi^2} \log L_h + O(1). \quad (101)$$

We do not interpret a TI Bogoliubov-mode occupation variance as a conserved physical $U(1)$ charge variance: the TI model does not have that number symmetry. In the XX model the entropy is nonlinear in v_{def} but the variance is linear, so their leading ratio is

$$\frac{a_{S_1}^{\text{XX}}}{a_F^{\text{XX}}} = \frac{4\mathcal{I}_1(s)}{s^2}. \quad (102)$$

It equals $\pi^2/3$ at $s = 1$, not at general transmission. A naive homogeneous conversion of the suppressed variance coefficient into a level would therefore measure transmission-weighted information, not a change of the bulk current algebra.

Finally change variables from ξ to the actual signed XX modular energy $\epsilon = \pm 2\Omega_s(\xi)$. The Jacobian gives, for $0 < s < 1$,

$$\begin{aligned} \varrho_s(\epsilon) &= \frac{\log L_h}{2\pi^2} \frac{s \sinh(|\epsilon|/2)}{\sqrt{s^2 \cosh^2(\epsilon/2) - 1}} \Theta_{\text{H}}(|\epsilon| - \epsilon_*) + \varrho_{\text{sub}}(\epsilon), \\ \epsilon_* &= 2 \operatorname{arcosh}(1/s). \end{aligned} \quad (103)$$

Here Θ_{H} is the Heaviside function. To check the normalization, use $d\xi/d|\epsilon| = s \sinh(|\epsilon|/2)/[2\sqrt{s^2 \cosh^2(\epsilon/2) - 1}]$ and count both signs once. The density has a modular gap and an integrable square-root threshold, becoming flat only at $s = 1$. Nevertheless its overall amplitude still grows as $\log L_h$. This explains how a gapped entanglement spectrum and a logarithmic entropy coexist, without gapping the physical critical chain.

The lesson for the adaptive-wall interpretation is a logical one. Before using a logarithmic coefficient as a CFT datum, one must specify the state, the entangling geometry, and any transmitting interface. The solvable defect supplies a counterexample to an inference, not evidence that the adaptive circuit has that defect spectrum or that its hard-wall support parameter equals s .

IX. WHAT CAN BE INFERRED ABOUT INDIVIDUAL WALLS?

A. One-wall and two-wall normalization

For a full-width strip $A(y_0, \ell) = [0, N_x) \times [y_0, y_0 + \ell)$ on a torus, both interfaces are intersected. The two endpoints of the interval are already included in the twist two-point function; one must not count them as an additional pair of wall branches. Table I collects the normalizations for charge-one complex fermions.

For two identical single-branch walls, extracting the per-wall quantities from full-strip slopes gives

$$c_w = \frac{6a_{S_q}}{1 + 1/q}, \quad k_w = \pi^2 a_F. \quad (104)$$

For a genuinely isolated single branch the conversion instead is $c_w = 12a_{S_q}/(1 + 1/q)$ and $k_w = 2\pi^2 a_F$. At $q = 1$, these are $3a_{S_1}$ for either wall of an identical pair and $6a_{S_1}$ for a single branch. In the usual nonchiral convention the Dirac theory is called $c = 1$ because $c_L = c_R = 1$; our explicit sum for that same theory is $c_\Sigma = 2$.

For a particular wall,

$$S_1^{(w)}(\ell) = \frac{c_L^{(w)} + c_R^{(w)}}{6} X_L(\ell) + \cdots, \quad c_-^{(w)} = c_R^{(w)} - c_L^{(w)}. \quad (105)$$

TABLE I. Leading coefficients of $X_L(\ell)$ for independent free branches. The last three columns are slopes, not the entropies or variances themselves. Opposite propagation orientations cancel signed central charge but not these positive coefficients.

Content	c_Σ	k_Σ	a_{S_q}	a_{S_1}	a_F
One chiral complex branch	1	1	$(1 + q^{-1})/12$	1/6	$1/(2\pi^2)$
Two opposite chiral walls	2	2	$(1 + q^{-1})/6$	1/3	$1/\pi^2$
One nonchiral Dirac channel	2	2	$(1 + q^{-1})/6$	1/3	$1/\pi^2$

Only if the wall is purely chiral does its entropy slope determine $|c_-^{(w)}|/6$. In a unitary theory, the general statement is merely $|c_-^{(w)}| \leq c_L^{(w)} + c_R^{(w)}$. Static entropy and charge fluctuations supply neither a propagation direction nor a proof that one chirality is absent. The explicit left/right replica weights in Ref. [13], Sec. II, make this distinction transparent.

B. Where chirality appears, and why the two-wall total cancels

Fix one common positive y -direction for the whole strip. For two oppositely propagating charge-one walls, the possible assignment is

$$(c_R^{(x_R)}, c_L^{(x_R)}) = (1, 0), \quad (c_R^{(x_L)}, c_L^{(x_L)}) = (0, 1). \quad (106)$$

Interchanging the wall labels or reversing y reverses the signed assignment. Each wall has $|c_-^{(w)}| = 1$, but the full pair has $c_R^{\text{tot}} = c_L^{\text{tot}} = 1$ and $c_-^{\text{tot}} = 0$. Its entropy nevertheless has slope $(c_R^{\text{tot}} + c_L^{\text{tot}})/6 = 1/3$. There is no contradiction: the entropy adds two positive contributions, while net chirality subtracts propagation directions. An alternative boundary-orientation convention reverses the tangent on the other edge; one must convert both edges to the same direction before taking a total.

The replica derivation contains this distinction before the equal-time limit is taken. With weights assigned to physical directions,

$$h_q^R + h_q^L = \frac{c_R + c_L}{24}(q - q^{-1}), \quad h_q^R - h_q^L = \frac{c_R - c_L}{24}(q - q^{-1}). \quad (107)$$

The first combination is the twist scaling dimension. The second is its signed spin (with the chosen orientation), which governs sensitivity to relative frame rotations or boosts rather than the ordinary equal-time chord length. This is how entanglement can carry information about a gravitational anomaly without the static coefficient itself being $c_R - c_L$ [13], Sec. II.2, Eqs. (21)–(22), and Sec. II.3.2.

The same sum-versus-difference distinction occurs for charge. To make the orientation explicit, expand the densities at equal time as

$$\rho_R(y) = \frac{1}{L} \sum_n J_n^R e^{2\pi i n y / L}, \quad \rho_L(y) = \frac{1}{L} \sum_n J_n^L e^{-2\pi i n y / L}. \quad (108)$$

Both mode algebras have the positive-level form of Eq. (58). Since $\partial_y \delta_L(y) = L^{-1} \sum_n (2\pi i n / L) e^{2\pi i n y / L}$, the total physical density has contact commutator

$$[\rho_R(y) + \rho_L(y), \rho_R(y') + \rho_L(y')] = -\frac{i(k_R - k_L)}{2\pi} \partial_y \delta_L(y - y') \hat{1}. \quad (109)$$

Its symmetric, separated-point fluctuations instead add the two levels, giving $a_F = (k_R + k_L)/(2\pi^2)$. Thus $k_- = k_R - k_L$ measures the signed charged anomaly, while $k_\Sigma = k_R + k_L$ measures the fluctuation strength. The difference and its global quantization conditions are discussed in Ref. [17], Secs. 2.1–2.2. An ordinary Dirac fermion has two individually level-one currents and zero k_- ; quantized levels do not establish net chirality. Moreover k_- and c_- remain different anomalies: neutral modes contribute to the latter without contributing to charge.

Accordingly, a full-strip entropy or variance fit can support the expected total sector content, but distinguishing separated chiral walls from a nonchiral channel requires spatial information and an independent direction-sensitive argument. An integer-looking coefficient alone does not supply that argument, and the defect example shows an additional reason not to identify every fitted logarithmic coefficient with a protected algebraic quantity.

C. Entanglement contours and wall windows

For a Gaussian reduced state define the cell-resolved contour [20]

$$s_q(\mathbf{r}; A) = \sum_{\mu} \langle \mathbf{r}, \mu | h_q(G_A) | \mathbf{r}, \mu \rangle = \sum_{\mu, a} |U_{\mathbf{r}\mu, a}|^2 h_q(\nu_a), \quad (110)$$

where $h_q(\nu)$ is the single-mode entropy kernel in Eq. (8). For $q = 1$ it is Eq. (9). The contour is nonnegative and obeys the exact sum rule

$$\sum_{\mathbf{r} \in A} s_q(\mathbf{r}; A) = S_q(A). \quad (111)$$

The diagonal is unchanged by transposing the matrix function, so this formula is consistent with Eq. (5). Likewise $f(\mathbf{r}; A) = \sum_{\mu} \langle \mathbf{r}, \mu | \mathcal{V}_A | \mathbf{r}, \mu \rangle$ sums to F_A .

For a window W_w surrounding one wall, define its assigned contribution

$$S_q^{[W_w]}(A) = \sum_{\mathbf{r} \in A \cap W_w} s_q(\mathbf{r}; A). \quad (112)$$

This is not generally $S_q(A \cap W_w)$. The former uses the matrix function of the original G_A ; the latter first restricts G to a different subsystem and then applies the nonlinear function. Restriction and taking the entropy kernel do not commute.

To interpret the coefficient of Eq. (112) as the contribution of an independent wall branch, the window must capture that branch's leading logarithmic weight; coupling to other walls must be negligible at the scales in question; and excluded tails must not carry an unaccounted logarithmic contribution. The contour sum rule alone proves none of these statements. Contours distribute a total entropy according to a specified prescription rather than defining a unique local observable. The interpretation of a finite window therefore needs a localization argument as well as the exact Gaussian formula.

Under those assumptions, one charge-one chiral wall contributes a von Neumann slope $a_{S_1}^{[W_w]} = 1/6$, or weight $3a_{S_1}^{[W_w]} = 1/2$ in the usual full-strip normalization. The two opposite walls contribute $1/2 + 1/2 = 1$ while their signed chiral central charges cancel. Without those assumptions, the window slope remains a well-defined assigned entropy coefficient, not automatically a central charge.

X. SCOPE OF THE DERIVATIONS

The correlation-spectrum expressions, counting determinant, cumulants, susceptibility identities, and contour closure are exact for the specified number-conserving Gaussian states. They apply before record averaging. The twist-field and current-correlator calculations instead assume the stated infrared CFT and its vacuum-like interval state, with the physical charge normalization held fixed. Passing from a full strip to independent wall contributions adds a localization and decoupling assumption.

The continuum actions specify useful comparison theories, not the adaptive evolution law. In particular a bosonically Gaussian Luttinger liquid is not necessarily a fermionically Gaussian state. The exactly solvable defect makes another distinction explicit: a critical bulk can retain its operator algebras while an interface changes the entropy coefficient and the modular spectral density. Neither the Luttinger parameter nor the defect transmission is an additional parameter of the adaptive protocol unless a separate mapping establishes it.

These distinctions explain both the usefulness and the limits of the comparison. Entropy and charge fluctuations probe different functions of one microscopic spectrum and different data of a continuum theory. Agreement with a charge-one complex-fermion prediction is informative, but it does not turn a static unsigned observable into a measurement of chirality or establish a preparation-depth bound.

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